HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed **RTL** for a **low-latency trading application** targeting high-frequency data processing, owning the design from specification through verification.
- Contributed to the improvement of the team's **build, simulation, and verification infrastructure**, enhancing code coverage and reducing simulation times by approximately **15%**.
- Coordinated and executed hardware deployments in **live production environments**, ensuring minimal disruption to ongoing trading operations and adhering to strict deployment windows.
- Developed and maintained **RTL designs** using **SystemVerilog** for complex digital logic within the firm's low-latency trading platform.
- Participated in **code reviews** for RTL designs, ensuring adherence to coding standards and best practices for maintainability and performance.
- Collaborated with cross-functional teams on the **specification and estimation** of new hardware features for the trading platform.
- Containerized development and testing environments using **Docker**, streamlining the setup process for new team members and ensuring consistent build outputs.

SKILLS

Hardware Design – RTL, FPGA, ASIC, SystemVerilog, VHDL, Xilinx/AMD, Synopsys, Cadence, AXI

Networking – Ethernet/MAC/PHY, PCIe, TCP/UDP, IP multicast, hardware protocol offloading

Programming – Python, C++, Shell Scripting

Tools & Infrastructure – Git, Jenkins, Docker, Kubernetes

Domain – Low-latency trading, Financial Markets, Hardware Deployments

PROJECTS

Low-Latency FPGA Trading Accelerator

- Designed and implemented **RTL logic in SystemVerilog** for an FPGA-based accelerator to process market data streams with microsecond latency.
- Developed a **simulation framework using cocotb** to rigorously test RTL designs, achieving over **95%** functional coverage on critical modules.
- Integrated the FPGA design with existing **AXI interfaces** for seamless data exchange with the host system and onboarded 2 engineers to the project's CI/CD pipeline.

Hardware Deployment Automation Framework

- Built a **Python-based automation framework** to streamline the deployment of FPGA configurations to production hardware, reducing deployment time by **40%**.
- Developed Jenkins pipelines to automate the build, test, and deployment stages of the FPGA design flow, integrating with **Git** for version control.
- Created comprehensive **runbooks and documentation** for the automated deployment process, enabling operations teams to execute deployments independently.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)